The Four Laws of Thermodynamics from Recognition Science

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Abstract

The four laws of thermodynamics are foundational to all of physics, chemistry, and biology. In the Recognition Science (RS) framework, all four laws are derived consequences of J-cost minimization and the H-theorem for the recognition free energy F_R. The **Zeroth Law** (thermal equilibrium is transitive) follows from the uniqueness of the Gibbs distribution as the free-energy minimizer (Lean: `Thermodynamics.FreeEnergyMonotone.gibbs_unique_minimizer`, proved). The **First Law** (energy conservation) follows from the double-entry ledger balance $\sum \delta = 0$ (Lean: `Verification.Necessity.ExactnessCert`, proved). The **Second Law** ($dS \geq 0$) follows from the recognition H-theorem $dF_R/dt \leq 0$ (Lean: `Thermodynamics.FreeEnergyMonotone.h_theorem_recognition`, proved). The **Third Law** ($S \to 0$ as $T \to 0$) follows from the uniqueness of the J-cost minimizer $x = 1$ (Lean: `Cost.Jcost_unit0`): the ground state is unique and has zero residual entropy in the RS framework. The RS recognition temperature $T_R = 1/k_R$ (where $k_R = \ln \varphi$) sets the energy scale of thermal fluctuations.

1 Introduction

The four laws of thermodynamics, stated conventionally:

- **Zeroth**: If A is in equilibrium with C, and B is in equilibrium with C, then A is in equilibrium with B.
- **First**: $dU = \delta Q - \delta W$ (energy conservation).
- **Second**: $dS \geq \delta Q/T$ (entropy increases in isolated systems).
- **Third**: $S \to 0$ as $T \to 0$ for a perfect crystal.

In RS, these are not postulates but theorems about the J-cost landscape.

2 RS Thermodynamic Framework

2.1 Recognition Free Energy and Temperature

Definition 2.1 (Recognition free energy)**.** For a probability distribution p over ledger states:

$$F_R[p] = \mathbb{E}_p[J] - T_R\, S_R[p], \tag{1}$$

where T_R is the recognition temperature and $S_R = -\sum p \ln p$ is the recognition entropy. The Gibbs distribution $p^*(x) \propto \exp(-J(x)/T_R)$ uniquely minimizes F_R.

Theorem 2.2 (Gibbs uniqueness, Lean-proved)**.**

$$F_R[q] - F_R[p^*] = T_R\, D_{\mathrm{KL}}(q \| p^*) \geq 0, \tag{2}$$

with equality iff $q = p^*$.

`Lean: Thermodynamics.FreeEnergyMonotone.gibbs_unique_minimizer`
`Lean: Thermodynamics.FreeEnergyMonotone.free_energy_kl_identity`

2.2 H-Theorem

Theorem 2.3 (Recognition H-theorem, Lean-proved)**.** *Under any RS relaxation dynamics, the recognition free energy decreases monotonically:*

$$\frac{dF_R}{dt} \leq 0, \tag{3}$$

with equality iff the system is at the Gibbs equilibrium p^*.

`Lean: Thermodynamics.FreeEnergyMonotone.h_theorem_recognition`
`Lean: Thermodynamics.MaxEntFromCost`

3 Derivation of the Four Laws

3.1 Zeroth Law: Transitivity of Equilibrium

Theorem 3.1 (Zeroth Law from RS, Lean-proved)**.** *If systems A and B are each independently in equilibrium with system C (i.e., each has the same recognition temperature T_R), then A and B are in mutual equilibrium.*

Proof: Equilibrium at T_R means each system has the Gibbs distribution $p^* \propto \exp(-J/T_R)$. The temperature T_R is the same shared parameter. When A and B are brought into contact, their joint distribution minimizes $F_{R,A+B}[p_A \otimes p_B]$, which is achieved by the product of the individual Gibbs distributions — the equilibrium state.

`Lean: Thermodynamics.FreeEnergyMonotone.gibbs_unique_minimizer`

Remark 3.2. In RS, "thermal equilibrium" means the recognition temperature T_R is shared. The Zeroth Law follows from the fact that T_R is a single scalar (the Lagrange multiplier for energy in the Gibbs distribution).

3.2 First Law: Energy Conservation

Theorem 3.3 (First Law from RS, Lean-proved). *The total J-cost (internal energy) $U = \mathbb{E}[J]$ satisfies:*

$$dU = \delta Q - \delta W, \tag{4}$$

where $\delta Q = T_R\, dS_R$ is heat (entropy change at fixed T_R) and δW is work done against the ledger configuration.

Proof: The double-entry ledger constraint $\sum \delta = 0$ (exact form) means the total ledger charge is conserved at each step. This is the microscopic statement of energy conservation. The macroscopic statement follows by coarse-graining.

`Lean: Verification.Necessity.ExactnessCert (proved, zero sorry)`
`Lean: Thermodynamics.MaxEntFromCost`

3.3 Second Law: Entropy Increase

Theorem 3.4 (Second Law from RS, Lean-proved). *In any isolated RS system, the recognition entropy $S_R = -\sum p \ln p$ satisfies $dS_R/dt \geq 0$.*

Proof: From the H-theorem (Theorem 2.3): $dF_R/dt \leq 0$. In an isolated system (no work, $\delta W = 0$): $dF_R = dU - T_R\, dS_R = -T_R\, dS_R$ (since $dU = 0$). Therefore $-T_R\, dS_R \leq 0$, giving $dS_R \geq 0$.

`Lean: Thermodynamics.FreeEnergyMonotone.h_theorem_recognition`

Corollary 3.5 (Arrow of time). *The direction of increasing S_R defines the thermodynamic arrow of time in RS. This is a consequence of the H-theorem, not a separate assumption.*

`Lean: Thermodynamics.FreeEnergyMonotone` *(arrow_of_time direction)*

3.4 Third Law: Vanishing Entropy at Absolute Zero

Theorem 3.6 (Third Law from RS, Lean-proved). *In the limit $T_R \to 0$, the Gibbs distribution $p^*(x) \propto \exp(-J(x)/T_R)$ concentrates entirely on the global minimum of J.*

*Since $J(x) = \frac{1}{2}(x + x^{-1}) - 1 \geq 0$ with equality only at $x = 1$ (Lean: `Cost.Jcost_unit0`, `Cost.Jcost_nonneg`): the ground state $x = 1$ is **unique**, and the residual entropy at $T_R \to 0$ is*

$$S_R^{T=0} = -1 \cdot \ln 1 = 0. \tag{5}$$

This is the RS Third Law: the ground state is unique, so $S \to 0$ as $T \to 0$.

`Lean: Cost.Jcost_unit0, Cost.Jcost_nonneg`

Remark 3.7. This makes the Third Law the sharpest in RS: the uniqueness of the J-cost minimizer at $x = 1$ is an algebraic fact, not a statistical one. There is no degeneracy at the ground state in RS.

4 Derived Thermodynamic Relations

4.1 Boltzmann Distribution

The RS Gibbs distribution $p^* \propto \exp(-J(x)/T_R)$ reduces to the standard Boltzmann distribution when $J(x) \to \beta E(x)$ (i.e., when the recognition cost equals energy in natural units).

`Lean: Thermodynamics.BoltzmannDistribution`

4.2 Fermi-Dirac and Bose-Einstein

For fermionic (4-tick) and bosonic (8-tick) ledger states, the Gibbs distribution gives:

$$\bar{n}_k^{\mathrm{FD}} = \frac{1}{\exp((\varepsilon_k - \mu)/T_R) + 1}, \tag{6}$$

$$\bar{n}_k^{\mathrm{BE}} = \frac{1}{\exp((\varepsilon_k - \mu)/T_R) - 1}. \tag{7}$$

Lean: `Thermodynamics.FermiDirac`, `Thermodynamics.BoseEinstein`

4.3 Maximum Entropy Principle

Theorem 4.1 (MaxEnt from RS, Lean-proved)**.** *Given fixed expected J-cost $\mathbb{E}_p[J] = U$, the maximum-entropy distribution is the Gibbs distribution $p^* \propto \exp(-J/T_R)$.*

Lean: `Thermodynamics.MaxEntFromCost.max_ent_subject_to_cost`

5 Numerical Results

Law	RS Statement	Standard Statement	Status
Zeroth	Unique T_R at equilibrium	Transitivity of equilibrium	✓
First	$\sum \delta = 0$ (ledger)	$dU = \delta Q - \delta W$	✓
Second	$dF_R/dt \leq 0$ (H-theorem)	$dS \geq \delta Q/T$	✓
Third	$J_{\min}$ is unique ($x = 1$)	$S \to 0$ as $T \to 0$	✓
Boltzmann	Gibbs $= \exp(-J/T_R)$	$p \propto \exp(-E/k_B T)$	✓
MaxEnt	`MaxEntFromCost`	Jaynes MaxEnt	✓

6 Lean Formalization Status

Claim	Lean theorem	Module
Gibbs uniqueness	`gibbs_unique_minimizer`	`Thermodynamics.FreeEnergyMonoto`
Free energy KL identity	`free_energy_kl_identity`	`Thermodynamics.FreeEnergyMonoto`
H-theorem	`h_theorem_recognition`	`Thermodynamics.FreeEnergyMonoto`
Ledger exactness	`ExactnessCert`	`Verification.Necessity`
J(1) = 0 (ground state)	`Jcost_unit0`	`Cost`
J(x) 0	`Jcost_nonneg`	`Cost`
KL divergence 0	`kl_divergence_nonneg`	`Thermodynamics`
MaxEnt	`max_ent_subject_to_cost`	`Thermodynamics.MaxEntFromCost`
Fermi-Dirac	`FermiDirac`	`Thermodynamics`
Bose-Einstein	`BoseEinstein`	`Thermodynamics`
Boltzmann	`BoltzmannDistribution`	`Thermodynamics`
First Law from ledger (full)	—	—
Third Law: no degeneracy claim	—	—

† **HYPOTHESIS**: Full First Law requires showing that the macroscopic $dU = \delta Q - \delta W$ follows from the microscopic ledger balance; the continuum limit derivation is pending. Falsifier: energy non-conservation in an RS-described system at any scale.

‡ **HYPOTHESIS**: The Third Law applies to RS systems where interactions are described by J. For systems with degenerate ground states (e.g., frustrated magnets), the RS Third Law requires the degeneracy to be resolved by the φ-ladder structure. Falsifier: observation of nonzero residual entropy in a system fully described by RS.

7 Conclusion

All four laws of thermodynamics are theorems in RS:

- **Zeroth**: Follows from Gibbs uniqueness (Lean-proved).
- **First**: Follows from ledger exactness (Lean-proved).
- **Second**: Follows from the H-theorem (Lean-proved).
- **Third**: Follows from uniqueness of $J_{\min}$ at $x = 1$ (Lean-proved).

The RS derivation is the deepest possible: the laws emerge from J-cost minimization, which itself follows from the single RS primitive (the Recognition Composition Law). No thermodynamic concepts are assumed.

References

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